Multiparticle Quantum Mechanics: a Mathematical Tour to Second Quantization

Su Youyou
Supervisor: Assoc. Prof. François Gay-Balmaz
School of Physical and Mathematical Sciences, NTU Singapore

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Abstract

When we first learn quantum mechanics, we study frameworks such as the Schrödinger equation, Heisenberg equation, and path integral formulation. In these cases, we often describe systems by assigning labels to individual particles; however, in practice, we cannot always distinguish each particle. For example, when we have two electrons, there is no physical meaning in specifying which electron is at which position. Therefore, describing the system by assigning individual labels to particles is not a valid approach. In this project, we explore how second quantization provides another way to describe systems with identical particles, and investigate the relation between quantum field theory and stochastic processes through the Onsager–Machlup approach.

1 Second Quantization

Definition 1.1 (Hilbert Space for a Single Particle) *Let the Hilbert space for a single particle be*

$$\mathcal{H}.$$

For example, for a single free particle

$$\mathcal{H} = L^2(\mathbb{R}^3),$$

where we can choose a set of orthogonal basis

$$\{|i\rangle\}_{i\in I}, \qquad \langle i|j\rangle = \delta_{ij}.$$

Definition 1.2 (Hilbert Space for Multiparticle) *For N distinguishable (for now) particles, the Hilbert space is the tensor product*

$$\mathcal{H}^{\otimes N} = \mathcal{H} \otimes \mathcal{H} \otimes \cdots \otimes \mathcal{H}$$

The basis can be written as

$$|i_1, i_2, \ldots, i_N\rangle = |i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_N\rangle$$

Note that for indistinguishable particles, we cannot use this.

Note that when $\mathcal{H} = L^2(\mathbb{R}^3)$, we have $\mathcal{H}^{\otimes N} \cong L^2((\mathbb{R}^2)^3)$. Therefore, a system with N particles can be described by

$$\Psi(\mathbf{x_1}, \mathbf{x_2}, \ldots, \mathbf{x_N})$$

Definition 1.3 (Permutation on Tensor Product) *Let π be any permutation on a group of N elements. For any*

$$\pi \in S_N,$$

define

$$U_\pi : \quad \mathcal{H}^{\otimes N} \to \mathcal{H}^{\otimes N}$$

as

$$U_\pi(\psi_1 \otimes \cdots \otimes \psi_N) = \psi_{\pi^{-1}(1)} \otimes \cdots \otimes \psi_{\pi^{-1}(N)}$$

Definition 1.4 (Bosonic Hilbert Space for N Particles) *It is defined such that swapping any particles gives the same space.*

$$\mathcal{H}_+^{(N)} = \{\Psi \in \mathcal{H}^{\otimes N} : U_\pi \Psi = \Psi \text{ for all } \pi \in S_N\}$$

Definition 1.5 (Fermionic Hilbert Space for N Particles) *It is defined such that swapping any even number of particles gives the same space while swapping odd number of particles gives the opposite of the space.*

$$\mathcal{H}_-^{(N)} = \{\Psi \in \mathcal{H}^{\otimes N} : U_\pi \Psi = \operatorname{sgn}(\pi)\Psi \text{ for all } \pi \in S_N\}$$

Definition 1.6 (Symmetrization Operator) *For a system of N particles, it is defined as*

$$P_+ = \frac{1}{N!} \sum_{\pi \in S_N} U_\pi.$$

It maps any N-particle system into a corresponding bosonic state.

Example 1.7 (Symmetrization Operator Acting on Two Particles) *For a system with two particles, the system is in the state $\psi \otimes \phi \in \mathcal{H} \otimes \mathcal{H}$.*

The symmetrization operator is

$$P_+ = \frac{1}{2!}(I + U_{12}).$$

Let it act on the system

$$P_+(\psi \otimes \phi) = \frac{1}{2}(\psi \otimes \phi + \phi \otimes \psi)$$

Notice that for any permutation $\pi \in S_N$, we have

$$U_\pi(P_+(\psi \otimes \phi)) = P_+(\psi \otimes \phi)$$

This shows that after acting the symmetrization operator, we get a bosonic state.

Definition 1.8 (Anti-Symmetrization Operator) *For a system of N particles, it is defined as*

$$P_- = \frac{1}{N!} \sum_{\pi \in S_N} \operatorname{sgn}(\pi) U_\pi.$$

It maps any N-particle system into a corresponding fermionic state.

Example 1.9 (Anti-Symmetrization Operator Acting on Two Particles) *For a system with two particles, the system is in the state $\psi \otimes \phi \in \mathcal{H} \otimes \mathcal{H}$.*

The anti-symmetrization operator is

$$P_- = \frac{1}{2!}(I - U_{12}).$$

Let it act on the system

$$P_-(\psi \otimes \phi) = \frac{1}{2}(\psi \otimes \phi - \phi \otimes \psi)$$

Notice that if we act I on it, it is not changed. If we swap the two particles,

$$U_{12}P_-(\psi \otimes \phi) = -\frac{1}{2}(\psi \otimes \phi - \phi \otimes \psi).$$

An extra minus sign shows up.

This shows that after acting the symmetrization operator, we get a fermionic state.

Corollary 1.10 (Pauli Exclusion Principle) *Suppose we have two fermions both in the state ψ. We have*

$$P_-(\psi \otimes \psi) = -\psi \otimes \psi$$

But

$$P_-(\psi \otimes \psi) = \frac{1}{2}(\psi \otimes \psi - \psi \otimes \psi) = 0$$

We see that this is not physically possible.

Definition 1.11 (Zero Sector) *It is defined as*

$$\mathcal{H}_\pm^{(0)} = \mathbb{C}.$$

It is a one-dimensional Hilbert space with the unit vector

$$\Omega.$$

Definition 1.12 (Bosonic Fock Space) *It is defined as*

$$\mathcal{F}_+(\mathcal{H}) = \bigoplus_{N=0}^{\infty} \mathcal{H}_+^{(N)}$$

Definition 1.13 (Fermionic Fock Space) *It is defined as*

$$\mathcal{F}_-(\mathcal{H}) = \bigoplus_{N=0}^{\infty} \mathcal{H}_-^{(N)}$$

Remark 1.14 (Why we need Fock space?) *Here, we don't know how many particles exactly we have, so we cannot decide which Hilbert space to use. Therefore, we direct sum up all Hilbert space and describe them in the Fock space. Note that the direct sum is actually a sequence of all Hilbert space of different number of particles.*

For example, a bosonic state could be

$$\Psi = (c_0, \Psi^{(1)}, \Psi^{(2)}, 0, 0, \dots)$$

where c_0 is the zero component, $\Psi^{(1)}$ is the one-particle component and $\Psi^{(2)}$ is the two-particle component.

It also has to be normalized

$$|c_0|^2 + \left|\Psi^{(1)}\right|^2 + \left|\Psi^{(2)}\right|^2 = 1$$

When we are measuring the number of particles, we have

$$P(N=0) = |c_0|^2$$

$$P(N=1) = \left|\Psi^{(1)}\right|^2$$

$$P(N=2) = \left|\Psi^{(2)}\right|^2$$

Example 1.15 (Two-Particle Fock Space) *For two bosons, swapping them gives the same state. So the wave function must satisfy*

$$\Psi(x_1, x_2) = \Psi(x_2, x_1)$$

All vectors satisfying this compose a subspace, denoted as

$$\mathcal{H}_+^{(2)} = \operatorname{Sym}^2 \mathcal{H}.$$

For two fermions, the sign changes after swapping two of them

$$\Psi(x_2, x_1) = -\Psi(x_1, x_2)$$

All vectors satisfying this compose a subspace, denoted as

$$\mathcal{H}_-^{(2)} = \bigwedge^2 \mathcal{H}$$

Definition 1.16 (Occupation Number States) *For indistinguishable particles, we cannot label the particles, so there is no way we can specify the state of a certain particle. So instead of telling the state of each particle, we use the number of particles occupying a certain eigenstate.*

For a single particle, the Hilbert space is $\mathcal{H}$. We take a set of basis $\{e_j\}_{j\in I}$. Each e_j is an eigenstate, which we call a one-particle mode.

For bosons, the occupation numbers satisfy

$$n_j \in \{0, 1, 2, \dots\} \qquad \sum_j n_j < \infty.$$

For fermions, the occupation numbers satisfy

$$n_j \in \{0, 1\}$$

The corresponding basis vector is

$$|n_1, n_2, \dots\rangle .$$

Each n_j describes the number of particles occupying the mode e_j.

The total number of particles is

$$N = \sum_j n_j$$

We have to note that the occupation number states depend on the basis we choose.

Definition 1.17 (Creation Operator) *For a chosen mode e_j, the creation operator is*

$$a_j^\dagger : \mathcal{F}_{+,fin}(\mathcal{H}) \to \mathcal{F}_{+,fin}(\mathcal{H})$$

where

$$a_j^\dagger |n_1, \ldots, n_j, \ldots\rangle = \sqrt{n_j + 1}\, |n_1, \ldots, n_j + 1, \ldots\rangle .$$

It adds the occupation number on e_j by 1.

Here $\mathcal{F}_{+,fin}$ is the Fock space of finite particles.

Definition 1.18 (Annihilation Operator) *For a chosen mode e_j, the annihilation operator is*

$$a_j : \mathcal{F}_{+,fin}(\mathcal{H}) \to \mathcal{F}_{+,fin}(\mathcal{H})$$

where

$$a_j |n_1, \ldots, n_j, \ldots\rangle = \sqrt{n_j}\, |n_1, \ldots, n_j - 1, \ldots\rangle .$$

It subtracts the occupation number on e_j by 1.

Specifically,

$$a_j \Omega = 0$$

Corollary 1.19 (Commutation Relation) *For bosons,*

$$[a_i, a_j] = [a_i^\dagger, a_j^\dagger] = 0,$$

$$[a_i, a_j^\dagger] = \delta_{ij}.$$

For fermions,

$$\{a_i, a_j\} = \{a_i^\dagger, a_j^\dagger\} = 0.$$

$$\{a_i, a_j^\dagger\} = \delta_{ij}.$$

Definition 1.20 (Number Operator) *For a single-particle state e_j, we define the number operator*

$$N_j = a_j^\dagger a_j.$$

For the entire system,

$$N = \sum_j a_j^\dagger a_j.$$

Definition 1.21 (Field Operator) *Recall that*

$$|x\rangle = \sum_i |i\rangle \langle i|x\rangle = \sum_i \phi_i^*(x)\, |i\rangle$$

where

$$\phi_i(x) = \langle x|i\rangle$$

describes the amplitude of finding the particle at position x while the wave function is in state $|i\rangle$.

The field creation operator that creates a particle at position x is defined as

$$\hat{\psi}^\dagger(x) = \sum_i \phi_i^*(x) a_i^\dagger.$$

Similarly, the field annihilation operator that annihilates a particle at position x is defined as

$$\hat{\psi}(x) = \sum_i \phi_i(x) a_i.$$

Example 1.22 (Create a Particle) *Suppose we have a vacuum state $|0\rangle$. Now we want to create one particle at position x,*

$$\hat{\psi}^\dagger(x)\,|0\rangle = \sum_i \phi_i^*(x) a_i^\dagger\,|0\rangle\,.$$

Here

$$a_i^\dagger\,|0\rangle = |0,\ldots,1,\ldots\rangle = |\phi_i\rangle\,.$$

Then

$$\hat{\psi}^\dagger(x)\,|0\rangle = \sum_i \phi_i^*(x)\,|i\rangle$$

Recall that

$$\phi_i^*(x) = \langle i|x\rangle$$

Therefore

$$\begin{aligned}\hat{\psi}^\dagger(x)\,|0\rangle &= \sum_i \langle i|x\rangle\,|i\rangle \\ &= \sum_i |i\rangle\,\langle i|x\rangle \\ &= |x\rangle\,.\end{aligned}$$

Now if we do it all again, we will get

$$\hat{\psi}^\dagger(x)\,|x\rangle = |x,x\rangle\,.$$

We have two particles at the position x.

Note that,

$$\mathcal{F} = \mathcal{H}_0 \oplus \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \ldots,$$

where

$$|0\rangle \in \mathcal{H}_0, \qquad |x\rangle \in \mathcal{H}_1, \qquad |x,x\rangle \in \mathcal{H}_2.$$

Remark 1.23 (Annihilate a Particle) *Suppose now we have a single particle state $|f\rangle$ with the wave function f. It can be written as*

$$|f\rangle = \int d^d y\,|y\rangle\,\langle y|f\rangle = \int d^d y f(y)\,|y\rangle\,.$$

Here, the d is the dimension of the Hilbert space and $d^d y$ is a short hand for multivariable integral. We also use

$$f(y) = \langle y|f\rangle\,,$$

which describes the amplitude of finding the particle at state y with wave function f.

Now that we know

$$|y\rangle = \hat{\psi}^\dagger(y)\,|0\rangle\,,$$

we can rewrite the single particle state

$$|f\rangle = \int d^d y f(y)\hat{\psi}^\dagger(y)\,|0\rangle$$

Then, we should apply the annihilation operator

$$\hat{\psi}(y)\left|f\right\rangle = \int d^d y f(y)\hat{\psi}(y)\hat{\psi}^\dagger(y)\left|0\right\rangle.$$

For bosons,

$$[\hat{\psi}(x), \hat{\psi}^\dagger(y)] = \delta^{(d)}(x-y).$$

And by the definition of annihilation operator,

$$\hat{\psi}(x)\left|0\right\rangle = 0.$$

Therefore, we have

$$\hat{\psi}(x)\left|f\right\rangle = \int d^d y f(y)\delta^{(d)}(x-y)\left|0\right\rangle = f(x)\left|0\right\rangle.$$

This shows that by annihilating a particle at position x, we get the amplitude of the function f at postion x.

Therefore, we have

$$f(x) = \langle 0|\hat{\psi(x)}|f\rangle.$$

Corollary 1.24 (Commutation Relation) *For bosons,*

$$[\hat{\psi}(x), \hat{\psi}^\dagger(y)] = \delta^{(d)}(x-y),$$
$$[\hat{\psi}(x), \hat{\psi}(y)] = 0,$$
$$[\hat{\psi}^\dagger(x), \hat{\psi}^\dagger(y)] = 0.$$

For fermions,

$$\{\hat{\psi}(x), \hat{\psi}^\dagger(y)\} = \delta^{(d)}(x-y),$$
$$\{\hat{\psi}(x), \hat{\psi}(y)\} = 0,$$
$$\{\hat{\psi}^\dagger(x), \hat{\psi}^\dagger(y)\} = 0.$$

Definition 1.25 (Number Density) *Define the number density as following*

$$\hat{n}(x) = \hat{\psi}^\dagger(x)\hat{\psi}(x).$$

It describes the density of particles near x.

Definition 1.26 (Number Operator) *Define the number operator as following*

$$\hat{N} = \int d^d x \hat{\psi}^\dagger(x)\hat{\psi}(x).$$

It describes the total number of particles.

Now, we first look at first quantization.

Suppose we have a system with a fixed number N particles. We can write the Hamiltonian in the following form.

$$\hat{H}_N = \sum_{i=1}^{N} \hat{h}_i + \sum_{i<j} U(x_i - x_j)$$

where

$$\hat{h}_i = -\frac{\hbar^2}{2m}\nabla_i^2 + V(x_i).$$

In this case, we give one label to each particle. But in practice, we cannot distinguish identical particles. So we need second quantization.

Example 1.27 (Second Quantization for a One-Body Hamiltonian) *The first-quantized one-body Hamiltonian is*

$$\hat{h}_i = -\frac{\hbar^2}{2m}\nabla_i^2 + V_i(x).$$

More specifically,

$$\hat{h}_i = I \otimes \cdots \otimes I \otimes h_i \otimes I \cdots \otimes I.$$

When it acts on N particles, it is

$$\hat{H}_N = \sum_{i=1}^{N} \hat{h}_i.$$

So after we take into consideration the inter-particle interaction, the first quantized Hamiltonian is

$$\hat{H}_N = \sum_{i=1}^{N} \hat{h}_i + \sum_{1\le i<j\le N} U(x_i - x_j)$$

Choose a set of orthogonal basis $\{|i\rangle\}_{i\in I}$ for a single particle in a Hilbert space. We have the following

$$\hat{h} = \sum_{i,j} h_{ij} \left|i\right\rangle\!\left\langle j\right|$$

where

$$h_{ij} = \langle i|\hat{h}|j\rangle .$$

Now this works only for one particle, we wish to have the Hamiltonian working for multiparticle thus on the Fock space. So we have the following

$$\hat{H}^{(1)} = \sum_{i,j} h_{ij} a_i^\dagger a_j.$$

We know that

$$\hat{I} = \int d^d x \left|x\right\rangle\!\left\langle x\right| ,$$

so we can rewrite h_{ij} as

$$\begin{aligned} h_{ij} &= \langle i| \left(\int d^d x \left|x\right\rangle\!\left\langle x\right| \right) \hat{h} \left(\int d^d x' \left|x'\right\rangle\!\left\langle x'\right| \right) |j\rangle \\ &= \int d^d x d^d x' \langle i|x\rangle \left\langle x\middle|\hat{h}\middle|x'\right\rangle \left\langle x'\middle|j\right\rangle . \end{aligned}$$

Substitute this back and we have

$$\begin{aligned} \hat{H}^{(1)} &= \sum_{i,j} \int d^d x d^d x' \langle i|x\rangle \left\langle x\middle|\hat{h}\middle|x'\right\rangle \left\langle x'\middle|j\right\rangle a_i^\dagger a_j \\ &= \int d^d x d^d x' \left[\sum_i \langle i|x\rangle a_i^\dagger \right] \left\langle x\middle|\hat{h}\middle|x'\right\rangle \left[\sum_j \left\langle x'\middle|j\right\rangle a_j \right] \\ &= \int d^d x d^d x' \hat{\psi}^\dagger(x) \left\langle x\middle|\hat{h}\middle|x'\right\rangle \hat{\psi}(x'). \end{aligned}$$

For the middle term, we have

$$\langle x|\hat{h}|x'\rangle = \langle x|\hat{T}|x'\rangle + \langle x|\hat{V}|x'\rangle$$

where

$$\langle x|\hat{T}|x'\rangle = -\frac{\hbar^2}{2m}\nabla_x^2\delta^{(d)}(x-x'),$$
$$\langle x|\hat{V}|x'\rangle = V(x)\delta^{(d)}(x-x').$$

Substitute it back and we get

$$\langle x|\hat{h}|x'\rangle = \left[-\frac{\hbar^2}{2m}\nabla_x^2 + V(x)\right]\delta^{(d)}(x-x').$$

Since

$$\int d^dx'\delta^{(d)}(x-x')\hat{\psi}(x') = \hat{\psi}(x),$$

we get the second-quantized one-body Hamiltonian

$$\hat{H}^{(1)} = \int d^dx\hat{\psi}^\dagger(x)\left[-\frac{\hbar^2}{2m}\nabla_x^2 + V(x)\right]\hat{\psi}(x).$$

Example 1.28 (Second Quantization for Two-Body Interaction) *Consider the first-quantized two-body interaction*

$$\hat{H}_N^{(2)} = \sum_{1\le a<b\le N} U(\hat{x}_a - \hat{x}_b).$$

Equivalently, this can be written as

$$\hat{H}_N^{(2)} = \frac{1}{2}\sum_{\substack{a,b=1\\a\neq b}}^{N} U(\hat{x}_a - \hat{x}_b),$$

where the factor $1/2$ removes the double counting of the particle pairs (a,b) and (b,a).

Let $\{|i\rangle\}_{i\in I}$ be an orthonormal basis of the single-particle Hilbert space. The corresponding basis of the two-particle Hilbert space is

$$|i,j\rangle = |i\rangle\otimes|j\rangle.$$

The two-particle interaction operator can therefore be expanded as

$$\hat{U} = \sum_{i,j,k,l} U_{ij;kl}\,|i,j\rangle\langle k,l|,$$

where

$$U_{ij;kl} = \langle i,j|\hat{U}|k,l\rangle.$$

The corresponding operator on Fock space is

$$\hat{H}^{(2)} = \frac{1}{2}\sum_{i,j,k,l} U_{ij;kl}a_i^\dagger a_j^\dagger a_l a_k.$$

Here, $a_l a_k$ annihilates two particles in the single-particle states $|k\rangle$ and $|l\rangle$, while $a_i^\dagger a_j^\dagger$ creates two particles in the states $|i\rangle$ and $|j\rangle$.

We now rewrite this operator in the position representation. Using the completeness relation

$$\hat{I} = \int d^d x \; |x\rangle\langle x| \,,$$

in each single-particle Hilbert space, we obtain

$$\begin{aligned} U_{ij;kl} &= \langle i,j|\hat{U}|k,l\rangle \\ &= \int d^d x \, d^d y \, d^d x' \, d^d y' \; \langle i|x\rangle \, \langle j|y\rangle \; \big\langle x,y\big|\hat{U}\big|x',y'\big\rangle \, \big\langle x'\big|k\big\rangle \, \big\langle y'\big|l\big\rangle \,. \end{aligned}$$

Substituting this expression into the second-quantized operator gives

$$\begin{aligned} \hat{H}^{(2)} &= \frac{1}{2} \sum_{i,j,k,l} \int d^d x \, d^d y \, d^d x' \, d^d y' \; \langle i|x\rangle \, \langle j|y\rangle \; \big\langle x,y\big|\hat{U}\big|x',y'\big\rangle \, \big\langle x'\big|k\big\rangle \, \big\langle y'\big|l\big\rangle \, a_i^\dagger a_j^\dagger a_l a_k \\ &= \frac{1}{2} \int d^d x \, d^d y \, d^d x' \, d^d y' \; \left[\sum_i \langle i|x\rangle \, a_i^\dagger\right] \left[\sum_j \langle j|y\rangle \, a_j^\dagger\right] \; \big\langle x,y\big|\hat{U}\big|x',y'\big\rangle \left[\sum_l \big\langle y'\big|l\big\rangle \, a_l\right] \left[\sum_k \big\langle x'\big|k\big\rangle \, a_k\right]. \end{aligned}$$

Using the definitions of the field operators

$$\hat{\psi}^\dagger(x) = \sum_i \langle i|x\rangle \, a_i^\dagger, \qquad \hat{\psi}(x) = \sum_i \langle x|i\rangle \, a_i,$$

we obtain

$$\hat{H}^{(2)} = \frac{1}{2} \int d^d x \, d^d y \, d^d x' \, d^d y' \; \hat{\psi}^\dagger(x) \hat{\psi}^\dagger(y) \, \big\langle x,y\big|\hat{U}\big|x',y'\big\rangle \, \hat{\psi}(y') \hat{\psi}(x').$$

For a local two-body interaction depending only on the relative position of the two particles, the position-space kernel is

$$\big\langle x,y\big|\hat{U}\big|x',y'\big\rangle = U(x-y)\delta^{(d)}(x-x')\delta^{(d)}(y-y').$$

Substituting this kernel gives

$$\begin{aligned} \hat{H}^{(2)} &= \frac{1}{2} \int d^d x \, d^d y \, d^d x' \, d^d y' \; \hat{\psi}^\dagger(x) \hat{\psi}^\dagger(y) U(x-y) \delta^{(d)}(x-x') \delta^{(d)}(y-y') \hat{\psi}(y') \hat{\psi}(x') \\ &= \frac{1}{2} \int d^d x \, d^d y \, U(x-y) \hat{\psi}^\dagger(x) \hat{\psi}^\dagger(y) \hat{\psi}(y) \hat{\psi}(x). \end{aligned}$$

Therefore, the second-quantized two-body interaction Hamiltonian is

$$\hat{H}^{(2)} = \frac{1}{2} \int d^d x \, d^d y \, U(x-y) \hat{\psi}^\dagger(x) \hat{\psi}^\dagger(y) \hat{\psi}(y) \hat{\psi}(x).$$

Remark 1.29 (Full Hamiltonian after Second Quantization) *Combine the two, we have the full Hamiltonian*

$$\hat{H} = \int d^d x \hat{\psi}^\dagger(x) \left[-\frac{\hbar^2}{2m}\nabla_x^2 + V(x)\right] \hat{\psi}(x) + \frac{1}{2} \int d^d x d^d y U(x-y) \hat{\psi}^\dagger(x) \hat{\psi}^\dagger(y) \hat{\psi}(y) \hat{\psi}(x).$$

Now that we have the formula of the second-quantized Hamiltonian, we have to show that the number of particles is constant as Hamiltonian evolving through time.

Heisenberg picture tells us that an operator follows

$$\frac{\mathrm{d}\hat{A}_H}{\mathrm{d}t} = \frac{i}{\hbar}[\hat{H}, \hat{A}_H] + \left(\frac{\partial \hat{A}}{\partial t}\right)_H.$$

Since

$$\left(\frac{\partial \hat{N}}{\partial t}\right)_H = 0,$$

we are only left to prove that

$$[\hat{H}, \hat{N}] = 0.$$

Remark 1.30 (Conservation of Particle Number) *We take bosons as an example.*

We first show that

$$[\hat{N}, \hat{\psi}^\dagger(x)] = \hat{\psi}^\dagger(x).$$

By the definition of number operator, we have

$$[\hat{N}, \hat{\psi}^\dagger(x)] = \int d^d y \left[\hat{\psi}^\dagger(y)\hat{\psi}(y), \hat{\psi}^\dagger(x)\right].$$

Using

$$[AB, C] = A[B, C] + [A, C]B,$$

we obtain

$$[\hat{N}, \hat{\psi}^\dagger(x)] = \int d^d y\, \hat{\psi}^\dagger(y) \left[\hat{\psi}(y), \hat{\psi}^\dagger(x)\right] + \int d^d y \left[\hat{\psi}^\dagger(y), \hat{\psi}^\dagger(x)\right] \hat{\psi}(y).$$

For bosons, we have commutation relation

$$\left[\hat{\psi}^\dagger(y), \hat{\psi}^\dagger(x)\right] = 0,$$

and

$$\left[\hat{\psi}(y), \hat{\psi}^\dagger(x)\right] = \delta^{(d)}(y - x),$$

Therefore,

$$[\hat{N}, \hat{\psi}^\dagger(x)] = \int d^d y\, \hat{\psi}^\dagger(y)\delta^{(d)}(y - x) = \hat{\psi}^\dagger(x).$$

Similarly, we have

$$[\hat{N}, \hat{\psi}(x)] = -\hat{\psi}(x).$$

Now, we will calculate

$$[\hat{N}, \hat{H}] = 0$$

by splitting it into two parts.

First, we will show the one-body term commutes with number operator.

$$\begin{aligned}
[\hat{N}, \hat{H}^{(1)}] &= \int d^d x \left[\hat{N}, \hat{\psi}^\dagger(x)\hat{h}\hat{\psi}(x)\right] \\
&= \int d^d x \left([\hat{N}, \hat{\psi}^\dagger(x)]\hat{h}\hat{\psi}(x) + \hat{\psi}^\dagger(x)\hat{h}[\hat{N}, \hat{\psi}(x)]\right) \\
&= \int d^d x \left(\hat{\psi}^\dagger(x)\hat{h}\hat{\psi}(x) + \hat{\psi}^\dagger(x)\hat{h}(-\hat{\psi}(x))\right) \\
&= 0
\end{aligned}$$

Then, we will show the two-body term commutes with the number operator. Since

$$\begin{aligned}\left[\hat{N}, \hat{\psi}^\dagger(x)\hat{\psi}^\dagger(y)\hat{\psi}(y)\hat{\psi}(x)\right] &= \hat{\psi}^\dagger(x)\hat{\psi}^\dagger(y)\hat{\psi}(y)\hat{\psi}(x) + \hat{\psi}^\dagger(x)\hat{\psi}^\dagger(y)\hat{\psi}(y)\hat{\psi}(x) \\ &\quad - \hat{\psi}^\dagger(x)\hat{\psi}^\dagger(y)\hat{\psi}(y)\hat{\psi}(x) - \hat{\psi}^\dagger(x)\hat{\psi}^\dagger(y)\hat{\psi}(y)\hat{\psi}(x) \\ &= 0.\end{aligned}$$

We have

$$[\hat{N}, \hat{H}^{(2)}] = 0.$$

Therefore,

$$[\hat{N}, \hat{H}] = [\hat{N}, \hat{H}^{(1)}] + [\hat{N}, \hat{H}^{(2)}] = 0.$$

This shows that, the Hamiltonian will not map a fixed-number state to another fixed-number state

$$e^{-i\hat{H}t/\hbar}\mathcal{H}^{(N)} \subset \mathcal{H}^{(N)}.$$

Definition 1.31 (Multiparticle Wave Function) *If a system has more than one particle, the wave function is in the following form*

$$\Psi(x_1, x_2, x_3, \ldots, x_N, t)$$

The meaning is the associated amplitude at time t and particle 1 at x_1, 2 at x_2 etc.. Therefore, the probability density is

$$|\Psi(x_1, x_2, x_3, \ldots, x_N, t)|^2$$

More specifically,

$$|\Psi(x_1, x_2, x_3, \ldots, x_N, t)|^2 dx_1 dx_2 \ldots dx_N$$

describes the probability at time t of particle 1 at $[x_1, x_1 + dx_1]$, 2 at $[x_2, x_2 + dx_2]$, ... , N at $[x_N, x_N + dx_N]$.

Postulate 1.32 (Multiparticle Schrödinger Equation) *A multiparticle system follows Schrödinger equation*

$$i\hbar\frac{\partial}{\partial t}\Psi(x_1, x_2, \ldots, x_N, t) = \hat{H}\Psi(x_1, x_2, \ldots, x_N, t)$$

An example for the Hamiltonian operator is

$$\hat{H} = \sum_{i=1}^{N}\left[-\frac{\hbar^2}{2m_i}\nabla_i^2 + V_{\text{ext}}(\mathbf{x}_i)\right] + \sum_{i<j} U(\mathbf{x}_i - \mathbf{x}_j)$$

a sum of the kinetic energy, the external potential energy and internal potential energy.

Definition 1.33 (Identical Particles) *For a system of two particles, if we have two distinguishable particles, for example an electron and a proton, the probability before and after exchange them should be different*

$$|\Psi(x_1, x_2)|^2 \neq |\Psi(x_2, x_1)|^2$$

But if we have two identical particles, the probability should be the same,

$$|\Psi(x_1, x_2)|^2 = |\Psi(x_2, x_1)|^2$$

which then gives us

$$\Psi(x_1, x_2) = \pm\Psi(x_2, x_1)$$

When the positive sign appears, it's called symmetric wave function and when the negative sign appears, it's called antisymmetric wave function.

Definition 1.34 (Bosons) *The wave function of bosons does not change before and after swapping two particles*

$$\Psi(\dots, x_i, \dots, x_f, \dots) = \Psi(\dots, x_f, \dots, x_i, \dots)$$

For example, photons are bosons.

Definition 1.35 (Fermions) *After swapping two particles, the sign of the wave function changes*

$$\Psi(\dots, x_i, \dots, x_f, \dots) = -\Psi(\dots, x_f, \dots, x_i, \dots)$$

If there are two fermions in the exact same state, swapping them does not change the wave function, then the property of fermions gives

$$\Psi = -\Psi$$

so

$$\Psi = 0$$

This is the Pauli exclusion principle: two fermions cannot stay in the same quantum state.

Some examples of fermions: electrons, protons, neutrons.

Theorem 1.36 (Multiparticle Heisenberg Equation) *For field operator, we apply the Heisenberg equation*

$$i\hbar \frac{\partial \hat{\psi}(x)}{\partial t} = [\hat{\psi}(x), \hat{H}].$$

To calculate, we split $\hat{H}$ into two parts: $\hat{H}^{(1)}$ and $\hat{H}^{(2)}$.

First, we derive the contribution from the one-body term. We have

$$\hat{H}^{(1)} = \sum_i E_i \hat{a}_i^\dagger \hat{a}_i.$$

Calculate the commutation relation

$$[\hat{a}_i, \hat{H}^{(1)}] = \sum_j E_j [\hat{a}_i, \hat{a}_j^\dagger \hat{a}_j].$$

Since

$$[\hat{a}_i, \hat{a_j}^\dagger] = \delta_{ij},$$

and

$$[\hat{a}_i, \hat{a}_j] = 0,$$

we have

$$i\hbar \frac{\mathrm{d}\hat{a}_j}{\mathrm{d}t} = E_j \hat{a}_j.$$

The solution is

$$\hat{a}_j(t) = \hat{a}_j(0) e^{-iE_j t/\hbar}.$$

Then, we calculate the contribution by the interacting term. We have

$$\hat{H}^{(2)} = \frac{1}{2} \sum_{klmn} U_{klmn} \hat{a}_k^\dagger \hat{a}_l^\dagger \hat{a}_n \hat{a}_m,$$

where $\{|u_j\rangle\}_{j\in I}$ *is an appropriate basis and*

$$U_{klmn} = \langle u_k, u_l|\hat{U}|u_m, u_n\rangle .$$

Calculate the commutation relation

$$[\hat{a}_j, \hat{H}^{(2)}] = \frac{1}{2}\sum_{klmn} U_{klmn}[\hat{a}_j, \hat{a}_k^\dagger \hat{a}_l^\dagger \hat{a}_n \hat{a}_m].$$

We can simplify the right hand side through the commutation relations and obtain

$$\left[\hat{a}_j, \hat{a}_k^\dagger \hat{a}_l^\dagger \hat{a}_m \hat{a}_n\right] = \delta_{jk}\, \hat{a}_l^\dagger \hat{a}_m \hat{a}_n + \delta_{jl}\, \hat{a}_k^\dagger \hat{a}_m \hat{a}_n.$$

Using this, we finally obtain

$$[\hat{a}_j, \hat{H}^{(2)}] = \sum_{klmn} U_{lmn} \hat{a}_l^\dagger \hat{a}_m \hat{a}_n.$$

Therefore, we have the second-quantized Heisenberg equation using the mode annihilation operator.

$$i\hbar \frac{\mathrm{d}\hat{a}_j}{\mathrm{d}t} = E_j \hat{a}_j + \sum_{klmn} U_{lmn} \hat{a}_l^\dagger \hat{a}_m \hat{a}_n.$$

Similar conclusion also holds for the mode creation operator.

Now we know that field operator is defined as the sum of the product between mode operator and basis function ϕ*. We multiply both side with* ϕ *and know how particles are created and annihilated through time evolution*

$$i\hbar \frac{\mathrm{d}\hat{\psi}(x,t)}{\mathrm{d}t} = \left(-\frac{\hbar^2}{2m}\nabla_x^2 + \hat{V}(x)\right)\hat{\psi}(x,t) + \int dy\, U(x-y)\hat{\psi}^\dagger(y,t)\hat{\psi}(y,t)\hat{\psi}(x,t).$$

Recall that the reason we do second quantization is that, instead of labeling the particles, we specify the number of particles on each quantum state. The entire system is now described on the Fock space, a direct sum of the original Hilbert space for each fixed number quantum state. We want to show that, the Schrödinger equation still holds after second quantization. To do so, we will extract the fixed-number state from the Fock space to the original N-particle Hilbert space.

Theorem 1.37 (Recover N-Particle Schrödinger Equation) *The Schrödinger equation in the Fock space is*

$$i\hbar \frac{\mathrm{d}\,|\Psi(t)\rangle}{\mathrm{d}t} = \hat{H}\,|\Psi(t)\rangle$$

where $\hat{H}$ *is the second-quantized Hamiltonian. For a quantum state in the Fock space, we define it as follows*

$$|\Psi(t)\rangle = \bigoplus_{N=0}^{\infty} |\Psi_N(t)\rangle .$$

We also have the relation between the N *particle state and the entire system*

$$\Psi_N(x_1, \ldots, x_N, t) = \frac{1}{\sqrt{N!}}\left\langle 0\middle|\hat{\psi}(x_N)\ldots\hat{\psi}(x_1)\middle|\hat{\Psi}(t)\right\rangle .$$

Extract the N*-particle state on both sides of the Schrd̈inger equation. For the left hand side we have*

$$i\hbar \frac{1}{\sqrt{N!}}\left\langle 0\middle|\hat{\psi}(x_N)\ldots\hat{\psi}(x_1)\frac{\mathrm{d}}{\mathrm{d}t}\middle|\hat{\Psi}(t)\right\rangle = i\hbar\frac{\partial}{\partial t}\Psi_N(x_1, \ldots, x_N, t).$$

The right hand side becomes

$$\frac{1}{\sqrt{N!}} \langle 0| \hat{\psi}(x_N) \cdots \hat{\psi}(x_1) \hat{H} |\Psi\rangle = \hat{H}_N \Psi_N(x_1, \ldots, x_N).$$

Combine them together, we recover the Schrödinger equation for N particles

$$i\hbar \frac{\partial}{\partial t} \Psi_N(t) = \hat{H}_N \Psi_N(t).$$

This shows that second quantization is a more generalized theory that already includes Schrödinger equation.

Theorem 1.38 (Quantum Field Path Integral) *In the ordinary path integral formalism, we cut time into slices and use completeness relation at each point to get the propagator. Here in quantum field path integral, we want to do the same thing. First, we introduce the coherent state.*

$$\hat{\psi}(x) |\phi\rangle = \phi(x) |\phi\rangle$$

The coherent state is introduced in order to convert the operator formulation of second quantization into a form suitable for path integrals. Since a coherent state satisfies

$$\hat{a}_i |\alpha\rangle = \alpha_i |\alpha\rangle ,$$

the annihilation and creation operators can be replaced, inside coherent-state matrix elements, by ordinary complex variables α_i and α_i^*. Moreover, coherent states satisfy a resolution of the identity, so they can be inserted between infinitesimal time-evolution operators in the same way that position eigenstates are inserted in the usual particle path integral. This allows the operator Hamiltonian $\hat{H}(\hat{a}^\dagger, \hat{a})$ to be rewritten as a complex-valued Hamiltonian function $H(\alpha^*, \alpha)$, leading naturally to a path integral over complex fields.

The completeness relation is

$$I = \int \mathcal{D}\phi^* \, \mathcal{D}\phi \, e^{-\int d^d x \, |\phi(x)|^2} \, |\phi\rangle \langle\phi| .$$

Following similar logic as in ordinary path integral formulation, we will have

$$K\left[\phi_f^*, \phi_i\right] = \int_{\phi(t_i)=\phi_i}^{\phi^*(t_f)=\phi_f^*} \mathcal{D}\phi^* \, \mathcal{D}\phi \, \exp\left[\frac{i}{\hbar} S[\phi^*, \phi]\right] ,$$

where the action functional is

$$S = \int dt \, d^d x \, \phi^*(x) \left(i\hbar\partial_t + \frac{\hbar^2}{2m}\nabla^2 - V \right) \phi(x) - \frac{1}{2} \int dt \, d^d x \, d^d y \, \phi^*(x)\phi^*(y) U(x-y) \phi(y)\phi(x).$$

Under this formulation, the path integral now instead of going over all possible paths $x_a(t)$, it integrate all possible field past

$$\phi(x, t).$$

Since the Fock space consists of Hilbert spaces of all possible number of particles, this path integral takes into account the contribution from all sectors such as zero sector, one-particle sector or any other sector.

Remark 1.39 (Recover N-Particle Path Integral) *Now that we have the quantum field path integral, we want to recover the ordinary path integral by projecting it onto the fixed-particle Hilbert space.*

First, we introduce the projection operator.

$$\hat{P}_N : \quad \mathcal{F} \to \mathcal{H}_N, \qquad \hat{P}_N = \frac{1}{2\pi}\int_0^{2\pi} d\theta\, e^{i\theta(\hat{N}-N)}.$$

We see that it has the property

$$\hat{P}_N\,|n\rangle = \frac{1}{2\pi}\int_0^{2\pi} d\theta\, e^{i\theta(n-N)}\,|n\rangle = \delta_{nN}\,|n\rangle\,.$$

It filters out the sector of certain number of particles. Therefore, inserting the projector into the quantum-field propagator gives

$$K_N = \langle X_f|\hat{P}_N e^{-\frac{i}{\hbar}\hat{H}T}\hat{P}_N|X_i\rangle.$$

Since

$$[\hat{H},\hat{N}] = 0,$$

the Hamiltonian preserves each particle-number sector,

$$[\hat{H},\hat{P}_N] = 0,$$

and therefore

$$K_N = \langle X_f|e^{-\frac{i}{\hbar}\hat{H}T}|X_i\rangle,$$

where both the initial and final states belong to $\mathcal{H}^{(N)}$.

The position states in this sector are created from the vacuum by the field creation operator:

$$|x_1,\ldots,x_N\rangle = \frac{1}{\sqrt{N!}}\hat{\psi}^\dagger(x_1)\ldots\hat{\psi}^\dagger(x_N)|0\rangle.$$

Thus the fixed-N propagator can be written directly in terms of the field operators:

$$K_N = \frac{1}{N!}\langle 0|\hat{\psi}(x_{Nf})\cdots\hat{\psi}(x_{1f})e^{-\frac{i}{\hbar}\hat{H}T}\hat{\psi}^\dagger(x_{1i})\cdots\hat{\psi}^\dagger(x_{Ni})|0\rangle$$

which is the N-particle component of the quantum-field propagator.

The corresponding path integral representation is obtained by restricting the field configurations to this sector. The resulting action is exactly the first-quantized N-particle action,

$$S_N = \int dt\left[\sum_{a=1}^{N}\frac{m}{2}\dot{x}_a^2 - \sum_{a=1}^{N}V(x_a) - \sum_{a<b}U(x_a-x_b)\right].$$

Therefore,

$$K_N = \int\prod_{a=1}^{N}\mathcal{D}x_a(t)\exp\left(\frac{i}{\hbar}S_N[x_1,\ldots,x_N]\right)$$

is recovered from the quantum-field path integral by projecting onto the fixed-particle-number subspace.

2 Analogy with Onsager-Machlup Approach to Langevin Equations

In the theory of physics, we often have everything deterministic, but in reality, we cannot take every factor into consideration. For example, in Brownian motion, we cannot calculate the motion of every water particle. Therefore, we introduce another term into the Newton's second law to describe the noise from the environment. This leads us to the Langevin equation.

Theorem 2.1 (Langevin Dynamics) *The general Langevin equation is*

$$m\ddot{x} = -\frac{\partial V}{\partial x} - \gamma\dot{x} + \eta(t),$$

where $-\frac{\partial V}{\partial x}$ *is the force from potential,* $-\gamma\dot{x}$ *is the damping force and* $\eta(t)$ *is random noise.*

When we study microscopic particles, we ignore the inertia term $m\ddot{x}$ *and obtain the overdamped Langevin equation*

$$\gamma\dot{x} = -\frac{\partial V}{\partial x} + \eta(t).$$

Here the noise satisfies

$$\langle \eta(t) \rangle = 0,$$

and

$$\left\langle \eta(t)\eta'(t) \right\rangle = 2D\delta(t - t').$$

D *is the diffusion constant.*

Without the loss of generality, we rewrite the equation

$$\eta(t) = \dot{x} - F(x).$$

Recall in path integral formulation where each trajectory is assigned to an amplitude, now we also want to have a similar mathematical structure. Each path is corresponding to a noise, therefore the probability of the noise is that of the path.

The probability of Gaussian noise is

$$P[\eta] \propto \exp\left(-\frac{1}{4D}\int dt\, \eta^2(t)\right).$$

Therefore, we have the probability of each path

$$P[x(t)] \propto \exp\left(-\frac{1}{4D}\int dt\, \big(\dot{x} - F(x)\big)^2\right).$$

To compare this with the Feymann's path integral, we define Onsager-Machlup action.

Definition 2.2 (Onsager-Machlup Action)

$$S_{\mathrm{OM}} = \frac{1}{4D}\int dt\, \big(\dot{x} - F(x)\big)^2.$$

After the definition of Onsager-Machlup action, we now have

$$P[x(t)] \sim e^{-S_{OM}}.$$

Now we can compare the two different formulation.
Feymann's path integral:

$$K = \int \mathcal{D}x\, e^{\,iS[x]/\hbar},$$

Onsager-Machlup approach:

$$P = \int \mathcal{D}x\, e^{-S_{\mathrm{OM}}[x]},$$

They both have the structure

$$\int \mathcal{D}x\, e^{-\text{action}}.$$

Example 2.3 (Harmonic Oscillator) *The potential for this system is*

$$V(x) = \frac{1}{2}kx^2.$$

The overdamped Langevin equation becomes

$$\dot{x} = -\lambda x + \eta(t),$$

where

$$\lambda = k/\gamma.$$

We calculate the Onsager-Machlup action

$$S_{\mathrm{OM}} = \frac{1}{4D}\int dt\,\left(\dot{x} + \lambda x\right)^2.$$

For each path $x(t)$, we have

$$P[x(t)] \propto \exp\left(-S_{\mathrm{OM}}[x]\right).$$

The deterministic trajectory

$$\dot{x} = -\lambda x$$

minimizes the action and is the most probable path under the noise. The noise term allows deviations from this trajectory, but trajectories requiring larger fluctuations have exponentially smaller probabilities.

Remark 2.4 (Parisi-Wu Quantization) *The analogy between stochastic dynamics and quantum path integrals can be further extended by the Parisi–Wu stochastic quantization. After Wick rotation, the Euclidean formulation of a quantum field theory is expressed as*

$$Z = \int \mathcal{D}\phi\, e^{-S_E[\phi]},$$

where $S_E[\phi]$ is the Euclidean action. The weight of each field configuration has the same mathematical form as a Boltzmann distribution in statistical mechanics.

Parisi and Wu proposed that this distribution can be obtained dynamically by introducing an additional fictitious time parameter τ, called stochastic time. The field configuration evolves according to a Langevin equation in the space of field configurations,

$$\frac{\partial \phi(x,\tau)}{\partial \tau} = -\frac{\delta S_E[\phi]}{\delta \phi(x,\tau)} + \eta(x,\tau),$$

where the first term drives the field towards configurations minimizing the Euclidean action, while $\eta(x,\tau)$ represents a Gaussian stochastic noise satisfying

$$\langle \eta(x,\tau) \rangle = 0,$$

and

$$\big\langle \eta(x,\tau)\eta(x',\tau') \big\rangle = 2D\,\delta(x-x')\delta(\tau-\tau').$$

This equation is the field-theoretic analogue of the Langevin equation discussed in the Onsager–Machlup approach. The deterministic relaxation competes with random fluctuations, and in the long stochastic-time limit the probability distribution of the field configurations approaches

$$P[\phi] \propto e^{-S_E[\phi]}.$$

Therefore, stochastic quantization provides an alternative interpretation of the Euclidean quantum field path integral: instead of directly summing over all field configurations, one may view the quantum measure as the equilibrium distribution of a stochastic process evolving in an auxiliary time.